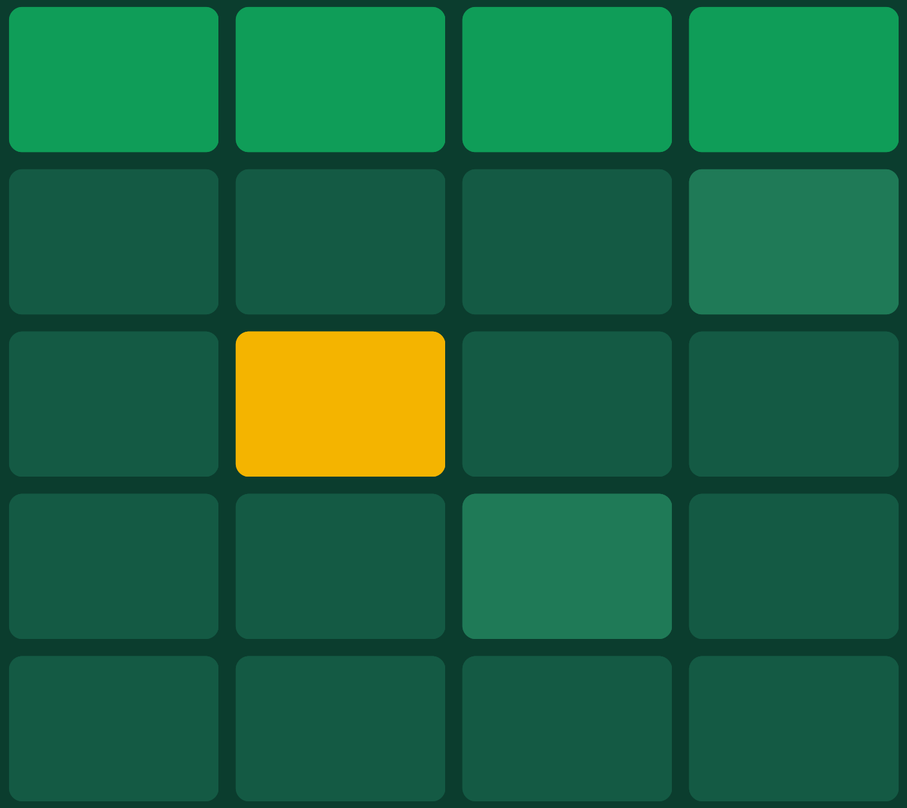
DISCUSSION

Google Sheets

A web-based spreadsheet for organizing, calculating, analyzing, and collaborating



PREPARED BY: GROUP 3

What is Google Sheets?

DEFINITION

A web-based spreadsheet application developed by Google.

Users create, organize, calculate, analyze, and present data using rows and columns.



Web-based

Accessed through the internet, unlike programs stored mainly on one computer.



Automatically saved

Changes are saved to Google Drive as you work.



Individual or group

Used alone or by multiple people at the same time, for personal and group work.

Purpose of Google Sheets

Organizing data

Arrange information systematically in rows and columns.

Performing calculations

Formulas and functions calculate totals, averages, and percentages.

Analyzing data

Sort, filter, compare, and summarize large amounts of information.

Presenting information

Convert data into charts and graphs that are easier to understand.

Collaborating with others

Multiple users work on the same spreadsheet simultaneously.

Storing information online

Files saved in Google Drive can be opened from different devices.

Interface of Google Sheets

Part	Description	Example
Rows	Horizontal sections identified by numbers	1, 2, 3, 4
Columns	Vertical sections identified by letters	A, B, C, D
Cell	Box formed by the intersection of a row and column	A1
Cell Reference	The address of a specific cell	B5
Formula Bar	Shows or edits the contents or formula of the selected cell	`=SUM(B2:B10)`
Sheet Tabs	Move between different sheets in one file	Sheet1, Sheet2
Toolbar	Tools for formatting and editing	Font, alignment, borders
Name Box	Shows the reference of the selected cell or range	A1
Grid	The rows and columns where data is entered	Spreadsheet area

Understanding Cell References

	A	B	C	D
1				
2				
3				
4		B4		
5				

Column B + Row 4 = cell B4

A cell reference identifies the exact location of data.

B = Column B

4 = Row 4

Therefore, B4 refers to the cell at the intersection of Column B and Row 4.

Collaboration & Sharing

Multiple people can work on the same spreadsheet at the same time and see each other's changes in real time. The owner controls access with three levels:



Viewer

Can only view the file



Commenter

Can view and add comments



Editor

Can make changes to the spreadsheet

This allows the owner to control who can access or modify the data.

Sorting & Filtering

Sorting

Organizes data according to a specific order. A list of students can be sorted:

- Alphabetically from A–Z
- Alphabetically from Z–A
- From lowest to highest score
- From highest to lowest score

Filtering

Temporarily displays only the information that meets a specific condition.

Example

A spreadsheet of 100 patients can be filtered to show only patients aged 60 and above. Other data is hidden temporarily, not deleted.

Charts & Graphs

Google Sheets converts data into visual representations that make results easier to understand.



Bar charts



Line graphs



Pie charts



Column charts



Area charts

Formatting & Validation

Conditional Formatting

Automatically changes the appearance of cells when they meet a specific condition.

Patient	Temperature (°C)
Patient A	36.8
Patient B	**38.4**
Patient C	37.1

Rule: Temperature ≥ 38°C → cell is highlighted

Data Validation

Controls what type of information can be entered into a cell, such as a dropdown list instead of typing a status.

Patient Status

Admitted
Discharged
Transferred

Helps reduce inconsistent data entry

Version History & Auto-Save

Version History

Keeps a record of changes made to the spreadsheet. Users can:

- See previous versions
- See who made changes
- See when changes were made
- Restore an earlier version

Automatic Saving

Changes are saved to Google Drive automatically, reducing the risk of losing work because of:

- Computer crashes
- Closing the browser accidentally
- Internet interruptions after changes have synced

Formulas & Functions

Formula

An expression used to perform a calculation. A formula usually begins with =

```
=A1+B1
```

Adds the values in A1 and B1

Function

A predefined formula that performs a specific calculation.

```
=SUM(B2:B10)
```

Adds the numbers in B2:B10

Common Functions

Function	Purpose	Example
SUM	Adds numbers	`=SUM(B2:B10)`
AVERAGE	Calculates the mean	`=AVERAGE(B2:B10)`
COUNT	Counts cells containing numbers	`=COUNT(B2:B10)`
MAX	Finds the highest value	`=MAX(B2:B10)`
MIN	Finds the lowest value	`=MIN(B2:B10)`
IF	Returns a result based on a condition	`=IF(B2>=75,"Passed","Failed")`
COUNTIF	Counts cells that meet a condition	`=COUNTIF(B2:B10,">=75")`
SUMIF	Adds values that meet a condition	`=SUMIF(A2:A10,"Nursing",B2:B10)`
ROUND	Rounds a number	`=ROUND(B2,2)`

Practical Use: Nursing



Patient Monitoring

- Vital signs
- Intake and output
- Blood glucose readings
- Patient schedules



Clinical Requirements

- Duty schedules
- Clinical hours
- Skills checklists
- Case assignments



Research

- Survey responses
- Participant data
- Descriptive statistics
- Tables and charts

More Practical Uses

Education

- Grade computation
- Attendance monitoring
- Class schedules
- Group projects
- Research data collection
- Survey results
- Assignment tracking

Personal Use

- Monthly budgeting
- Expense tracking
- Workout logs
- Meal planning
- Personal schedules
- To-do lists
- Savings goals

Workplace / Business

- Employee schedules
- Inventory tracking
- Sales records
- Budgeting
- Project management
- Attendance
- Task assignments
- Reports

Personal example: **=SUM(B2:B31)** totals a month of daily expenses automatically.

Key Takeaways



Organize

Arrange data in rows and columns



Calculate

Formulas and functions give automatic results



Analyze

Sort, filter, and visualize with charts



Collaborate

Work together in real time, saved to Drive

Google Sheets makes data organization, computation, analysis, and collaboration easier.